

Qualia Lab

A general laboratory for experience, representation, and behavior

Live app: <https://qualialab.fueldeskpro.com>

Repository: <https://github.com/merolaagi/qualialab>

This record preserves the original iterations and adds the real Monty extension in section 17. Original browser-only runtime descriptions apply to the first four panels; the new Sensorimotor lab runs on the Mac mini. See DEPLOYMENT.md for publishing and service management.

Consolidated reconstruction of iterations 1-4, extended from redness to five sensory domains. Prepared from the recovered conversation "Comparing Redness Qualia" and the user's subsequent request for a more general app.

1. Source status and preservation

The conversation was retrieved through both available pages: 18 turns, from the opening consciousness discussion through the request for one app and a master document. Appendix A preserves the text of every returned user and assistant message in chronological order, including short voice-transcription fragments and incomplete sentences. It is the recovered textual record, not a reconstruction of missing audio.

The older generated apps, README files, and PDFs were referenced by content markers, but the retrieval returned no attachments, artifact payloads, or usable download links. Consequently, this is a new implementation of the functionality described in the conversation. It is not a source-code merge, and pixel-for-pixel or algorithm-for-algorithm equivalence with those missing artifacts cannot be verified. Content-reference markers in the transcript are replaced by an explicit unavailable-artifact notice; no missing artifact is silently presented as recovered.

This README and the companion PDF contain the same research narrative and recovered transcript. The generalized sensory model, precise equations, training procedure, metrics, and interface are implementation choices of this consolidation. They should not be attributed to the earlier prototypes unless explicitly supported by the transcript.

2. Quick start

Open `dist/index.html` in a modern browser. The downloadable app requires no installation, build, API key, or backend. Keep `engine.js`, `app.js`, `style.css`, `README.md`, and `Qualia-Lab-Research.pdf` beside `index.html`. You may also use the hosted app. Computation runs in your browser; a hosted site's access controls are separate from the experiment.

The display fonts may be fetched from Google Fonts when online; system sans-serif fallbacks work offline. Experiments, training, search, and session export work offline. No microphone, camera, scent device, thermal device, or calibrated sensory hardware is used. Sound is an abstract input, not played audio; touch, taste, and smell are not physically delivered stimuli.

Start in Experiment studio. Choose Vision and the Red apple preset. Both transparent-model agents start identical, so their internal distance is zero. Change the preset to Red ball to vary object identity while preserving the initial color controls. Expand the input sections to manipulate lighting, texture, other signals, context, memory associations, and emotion. Expand either agent's History & interventions to change only that agent.

Use the five modality buttons to choose which categorical report is observed. All channels remain part of the common input; changing the selected report does not erase other modalities. Visit Learning & alignment to train the two networks. Visit Behavioral twins to search for equal report/action cases with large hidden-state differences. The Research notebook summarizes the evolution and links these documents.

Export session saves the full input, both agent settings, learned weights when present, memory traces, sequence log, and current derived data as JSON. Import validates a session before accepting it, restores its core experiment state, and invalidates alignment/search results for recomputation. Files stay on your device; the app does not automatically save on refresh. Export before closing if you want to preserve your work.

3. The original question: red apple versus red ball

The discussion began with a comparison between the redness of an apple and a ball. This asks several different questions at once: whether their reflected light is the same, whether an observer assigns the same color category, whether the surrounding experience is the same, and whether the felt character of redness itself is the same. The app separates these questions as far as a toy model can.

The user's central intuition was that the same light should produce the same redness. That became a controlled experiment: hold a color-related input fixed while varying object identity and the circumstances around it. In this implementation, apple and ball are numeric object codes, -1 and +1. They are not rendered physical objects or models of surface reflectance. Identical sliders therefore mean identical selected model inputs, not proof of physically identical light at a retina.

The early assistant described apples as deeper or waxier and balls as more uniform or synthetic. Those were illustrative possibilities, not necessary differences or measurements. A waxy ball and a uniform apple are possible. The scientifically useful move is to control texture, lighting, object identity, and associations independently instead of assuming those properties follow from an object name.

The next move was from two objects to two observers: if both say "red," are their experiences the same? Agreement on a word does not uniquely specify their internal state. It also does not establish that their experiences differ. The experiment explores this underdetermination through inspectable computational states, without claiming access to another subject's feeling.

4. Identical observers, randomization, and the operational boundary

The user proposed two beings identical in every respect, receiving the same stimulus. In a deterministic program, identical parameters, inputs, history, and memory lead to identical outputs and hidden states. This is a control and follows from the computational setup. Whether complete physical identity also fixes phenomenal experience is a further philosophical question, not something this code settles.

The conversation then introduced randomized histories and systematic additions and subtractions of experience. Randomization supplies varied conditions; it does not create subjectivity. A useful research program asks which changes affect reports, actions, memory, or future responses, and which do not affect the observables under test.

An early suggestion called an internal variable Q the agent's "red experience state." The later conversation corrected this: naming a variable qualia does not make it a measure of experience. This consolidation consistently calls such quantities internal representations. Qualia refers to felt or phenomenal character in the philosophical discussion; no numeric field in the app measures it.

The assistant's early use of "my red" in the transcript is conversational language, not evidence that an AI assistant has color experience. Its hypothetical 650 nm stimulus also does not describe this app's input: the app uses abstract normalized channels rather than a spectral or retinal model.

5. Hypotheses and what would count as evidence here

H1 - Complete-state identity: identical deterministic agents, with identical weights, gates, histories, sequence traces, and input, should yield identical hidden vectors and behavior. Nonzero differences in this control indicate a bug, a missing control, or an unrecorded difference.

H2 - Contextual modulation: changing context, association, emotion, sensory channels, or temporal history can change a representation even when the shared external stimulus stays fixed. Evidence consists of reproducible changes under a single controlled intervention. This is a statement about the implemented model.

H3 - Coarse behavioral underdetermination: different hidden vectors can produce the same reported category and the same binary action. Evidence is a reproducible candidate meeting the declared matching criterion. It does not show equality of all possible behavior.

H4 - Coordinate ambiguity: some raw differences can be removed by a mapping between hidden coordinate systems. Evidence is lower error on held-out paired stimuli after fitting a mapping on a separate calibration sample. Low error under a flexible linear mapping is limited evidence about transformability, not proof of equal geometry or consciousness.

H5 - History-dependent learning: under the same architecture, initialization, training budget, and target definitions, different training correlations can lead to different learned states. Repeat seeds and compare held-out performance before generalizing. The current app supports reproducible seeds but does not compute confidence intervals or a population-level conclusion automatically.

H6 - Ingredient sensitivity: removing an input group may make two representations converge or diverge. Compare each intervention against the same frozen baseline. This reveals sensitivity of this model under channel zeroing; it is not a causal finding about human consciousness.

6. Iteration history and preserved capabilities

Iteration 1 - Exploratory controls. The first prototype described two subjects and adjustable redness, saturation, illumination, shadow, context, prior memory, emotion, and influence from other senses. It offered identical agents, randomized B context with a fixed stimulus, and an inverted private color-label mapping. The consolidation preserves these as shared controls, separate agent settings, reset/randomization, and explicit report-label swapping.

Iteration 2 - Explicit processing and intervention. The pipeline became stimulus -> sensory processing -> context -> memory -> internal representation -> report/behavior. Users could subtract memory, context, senses, and emotion, randomize B's history, or invert B's internal color mapping. The consolidation preserves a transparent hand-authored model, per-agent gates, paired ablations, and a hidden-unit inversion with an optional compensating readout.

Iteration 3 - Learned representations. Eight-dimensional hidden states replaced a prescribed redness formula. The described experiments were identical histories, different histories with the same color categories, and inverted red/green training labels. Alignment was introduced to distinguish raw coordinate differences from other representational differences. The consolidation includes actual trained hidden weights, these three history protocols, and a calibration/held-out ridge-alignment experiment.

Iteration 4 - Automated search and temporal history. The described app searched 1,000-15,000 conditions for matching report/action with maximal internal divergence within the sample. It added immediately preceding experience and ablation of memory, context, other senses, and shape. The consolidation includes those sample sizes, deterministic search, a temporal state update, loading the winning case, and paired ablations, plus emotion ablation and a stricter probability-matching option.

Generalized edition - The user's follow-up requested a high-end app with a general scope. Vision, sound, touch, taste, and smell now use one engine, one pair of agents, and one experimental workflow. This is an extension made during consolidation, not a claim that all five domains existed in the unavailable original artifacts.

Iteration 5 remains a proposed research direction. The earlier conversation suggested comparing entire representational geometries. This app does not claim to implement that full research program: its alignment test is inherited from iteration 3. Geometry-wide analyses, richer behavior batteries, uncertainty estimates, and systematic repeated-seed studies remain future work.

7. Generalized experience model

Each experience contains 16 normalized channels: red/green signal; saturation; illumination; shadow; light angle; object identity; pitch; loudness; texture; temperature; taste; smell; context; prior association; emotion; and previous-state trace. Saturation, illumination, shadow, and loudness lie in $[0,1]$. The remaining channels lie in $[-1,1]$. These are chosen abstractions, not calibrated units.

Vision reports green / neutral / red. Sound reports low tone / quiet / high tone. Touch reports cool / neutral / warm. Taste reports bitter / neutral / sweet. Smell reports earthy / neutral / floral. These category axes are deliberately narrow and do not span all possible experiences in a modality. For example, the touch report concerns temperature while texture can influence the hidden state.

Agent history settings add clipped offsets to context, association, and emotion. A sensory gain scales the six nonvisual channels (pitch, loudness, texture, temperature, taste, smell), including the selected channel when the active domain is nonvisual. The nonvisual-senses gate zeros all six channels. It should not be interpreted as preserving the active nonvisual stimulus. Individual raw sliders provide more selective interventions.

The shared red/green channel is transformed using saturation, illumination, shadow, and angle: $\text{effective color} = \text{color} \times (0.25 + 0.75 \text{ saturation}) \times (0.45 + 0.55 \text{ illumination}) \times (1 - 0.55 \text{ shadow}) \times (1 - 0.15 |\text{angle}|)$, clipped to $[-1, 1]$. This is a transparent toy formula, not color science or a spectral model. It makes contextual experiments legible while keeping the scientific boundary explicit.

The app distinguishes a raw shared input from the agent's processed input. Equal raw stimulus does not imply equal processed input when agent history or interventions differ. Sequence memories can also differ. All these differences should be disclosed when interpreting a comparison.

8. Transparent model and inversion controls

The hand-authored model produces eight tanh-bounded features combining effective color and context, object identity and texture, pitch and loudness, temperature and texture, taste and scent, context and emotion, association and temporal trace, and emotion and scent. These combinations are fixed by the programmer. They demonstrate mechanisms and controls; they do not discover the structure of experience.

Reports are read from these features through simple softmax heads. Approach/avoid is another softmax readout of an explicitly chosen valence combination. These output definitions are intentionally inspectable but arbitrary. The learned model uses the same output vocabulary with trained weights.

Three kinds of inversion must not be conflated. Report endpoint swapping changes the displayed categorical mapping and swaps endpoint probabilities in the active modality. Inverted red/green training labels instead changes B's supervised color targets during training. Hidden-unit inversion negates B's first hidden component at evaluation. None establishes an inverted phenomenal spectrum.

With Compensate the readout enabled, the readout reverses that hidden sign change before producing outputs. This gives a known same-behavior/different-coordinate control by construction. Turn compensation off to see whether the changed unit affects the report or action. In the learned model, the first hidden unit has no guaranteed color meaning, so its inversion is not called a learned redness inversion.

9. Learning implementation and reproducibility

Each agent is a 16-input, 8-hidden-unit tanh network. Five categorical heads each have three outputs, and one action head has two outputs, for 17 output logits. A second, 16-output tanh decoder reconstructs the input. All hidden weights, head weights, decoder weights, and biases are updated by stochastic gradient descent.

The objective is the sum of six categorical cross-entropies plus 0.1 times the sum of squared reconstruction errors. The implementation uses a fixed learning rate of 0.025, 700 examples per agent, and a selectable 12, 24, or 48 epochs. The same dataset is reused in a fixed order each epoch. There is no pretrained model, outside AI service, or hidden API call.

Canonical targets use simple thresholds: effective color, temperature, taste, and smell use boundaries at -0.22 and +0.22. Sound is quiet below loudness 0.12; otherwise the sign of pitch chooses low/high. The action target is approach when $0.5 \text{ emotion} + 0.25 \text{ association} + 0.15 \text{ context} + 0.1 \text{ taste}$ is nonnegative, otherwise avoid. These are imposed task rules, not discovered meanings.

Identical histories: the two networks begin with identical seeded parameters and see identical samples and targets in identical order. Exact equality is expected. Different associations: the same pseudorandom base draws are used, but B's context correlates negatively with color, association positively with pitch, emotion negatively with taste, and texture negatively with object code. The same target-generating rules apply to the transformed input. Thus action-label frequencies can change as a consequence of changed features; this is not a perfectly matched behavioral training distribution.

Inverted labels: B sees the same inputs, but the green/red color targets are swapped, leaving the middle category and other target rules unchanged. The separate UI report-label swap is not automatically applied. Treat the two interventions independently.

Training resets manual agent interventions and sequence memory, so the chosen protocol has a controlled starting point. Reset identical copies A's learned weights to B if learned models exist; it does not train both from scratch. Fresh common-distribution accuracy is measured on 300 examples for the currently selected domain with canonical labels. For inverted-label B, canonical color accuracy is expected to penalize the intentional mapping change. Training accuracy is not guaranteed and should be inspected.

The seeded generator, fixed sample order, and exported parameters support reproducibility within the same implementation. Floating-point rounding can vary slightly across browser engines. Record the seed, history, epochs, modality, model type, agent interventions, and temporal state with each result. Do not compare different app versions as if they had identical semantics.

10. Temporal memory and frozen comparisons

The Previous-state trace input is combined with a per-agent sequence memory before evaluation. Present experience records the current result and then updates each agent's memory: $\text{next memory} = \text{clip}(0.65 \times \text{current memory} + 0.35 \times \text{hidden unit 7})$. The preview then shows the current raw stimulus under the updated memory. The recorded sequence entry refers to the result before that update.

This update is a chosen, inspectable recurrence, not a trained recurrent network or a biological memory model. Unit 7 in the learned network has no guaranteed memory interpretation. Repeatedly presenting a stimulus can alter its subsequent state because the recurrence feeds back an internal component. Moving sliders alone does not advance the sequence.

Clear sequence resets both sequence traces and the displayed sequence log. It leaves the separate Prior association input and manual Memory association offsets in place. To remove all memory inputs, use the memory gate or memory ablation, which zeros association and previous-state channels.

Search and ablation freeze sequence memories. Search samples a shared previous-state input per candidate and adds each agent's fixed sequence trace. It does not mutate the session by presenting candidates. This makes candidate ranking comparable, while allowing differences in retained history to affect the results.

11. Metrics, alignment, and behavioral-twin search

Raw RMS distance is $\sqrt{\text{mean}((hA - hB)^2)}$ across eight hidden components. Its range is 0-2 for tanh states. Zero means coordinate equality in this model at this input. Cosine similarity measures orientation, not magnitude. If either vector is zero, the implementation returns zero by convention, and the cosine should not be treated as informative.

The coarse match criterion requires the same selected-domain report and the same approach/avoid action. The strict option additionally requires every probability in the selected report head and the approach probability to differ by at most 0.05. The other four report heads are not part of either criterion. Neither criterion establishes complete behavioral equivalence or equality over future sequences.

Search draws 1,000, 5,000, or 15,000 candidate experiences from the common synthetic generator with seed = session seed + 7103. Among matches it retains the case with the largest selected RMS metric. Raw ranking is coordinate-sensitive. Aligned ranking uses the current calibration map and may be selected only after alignment. The winner is a sample maximum, not a globally optimal or statistically certified maximum. No matches is a valid result.

Alignment uses 192 paired calibration inputs and 192 different evaluation inputs from a seeded common distribution. A ridge least-squares affine map fits B's eight components plus an intercept to A's eight components. Ridge regularization is 0.00001 on the normal-equation diagonal, including the intercept. Reported raw and aligned errors are square roots of mean per-example squared RMS errors on the held-out set. Calibration uses fresh synthetic previous-state inputs, without adding current sequence memory; using the map for a sequence-influenced search is therefore an extrapolation worth checking.

The map can rescale and shear, so it is not restricted to orthogonal coordinate changes and does not preserve distances in general. Low aligned error is evidence of predictability under this map on the tested distribution. High aligned error does not rule out nonlinear or other mappings. Neither result establishes phenomenal equivalence. Settings changes invalidate the map so stale calibrations are not silently reused.

Research on comparing neural representations motivates paying attention to invariances and the distribution used for comparisons. Kornblith et al. (2019), Similarity of Neural Network Representations Revisited, introduces CKA and discusses limitations of similarity measures. CKA is a future option here, not a metric implemented in this app. Source: <https://proceedings.mlr.press/v97/kornblith19a.html>

12. Ablation protocol

Run ablations evaluates the current shared raw input with the same two models and frozen memories. It then zeros memory (association and prior-state channels), context, the six nonvisual sensory channels, object shape, or emotion for both agents, one group at a time. An all-five-groups condition is also shown. The baseline has no additional zeros beyond the current manual gates.

Every result displays raw hidden RMS distance, whether the selected report/action still match, and change in distance from baseline. A negative delta means convergence in raw coordinates. A positive delta means divergence. Existing gates remain applied, so ablating an already-disabled group should have no extra effect. No retraining occurs.

Zero is an explicit neutralized code, not literal absence of a sensory apparatus. Such inputs may be outside a network's training distribution. Interactions can mean that separate ablation effects do not add to the combined effect. These limits matter particularly when claiming that a single ingredient explains a difference.

13. Suggested experiments

Experiment A - Identity control. Reset identical, keep all controls fixed, and compare both the report/action and all eight components. Repeat after training identical histories. Expected: zero raw distance. If you then give one agent different memories or gates, it is no longer a complete-state identity control.

Experiment B - Apple versus ball. Use the initial red apple, record/export, then use red ball with the same color controls. Compare the object-sensitive features. The preset restores default context and other input controls, so for a strict one-factor experiment after custom edits change only the Object slider instead of selecting a preset.

Experiment C - Context with stimulus fixed. Keep shared inputs fixed, change only B's context offset, and observe representation, report confidence, and action. Return the offset to baseline before testing B's emotion or association separately.

Experiment D - Known coordinate change. Reset identical, enable B's hidden-unit inversion, and keep compensation on. A nonzero raw distance with identical outputs is expected whenever the inverted component is nonzero. Alignment should substantially remove a consistent sign-remapping difference. This validates the coordinate caution; it is not discovery of private feelings.

Experiment E - Learned history. Train identical histories, record the control, then train different associations using the same seed and budget. Compare common-stimulus states, fresh accuracy, held-out alignment errors, and a behavioral-twin search. Repeat other seeds manually before making a broad claim.

Experiment F - Language mapping. Train inverted labels and examine red and green stimuli. Separate category disagreement induced by training from the UI's post-readout label swap. Try sound afterward to show that the training intervention specifically concerned vision labels.

Experiment G - Temporal dependence. Present a strongly emotional or memory-associated stimulus, then switch to a fixed test stimulus. Compare with the same test after Clear sequence. Export both sessions. Use memory ablation to test whether the difference depends on the retained trace.

Experiment H - Search and explain. Train different associations or change B's history; search 5,000 cases with coarse matching; inspect probability gap; try strict matching; load the winner; and run paired ablations. A useful report names the criterion, number of candidates, seed, rank metric, and any outputs that still differ.

Experiment I - Generalization across domains. Repeat a controlled comparison in sound, touch, taste, and smell. The selected report head changes; the common latent state can integrate all channels. Do not interpret successful model behavior as a psychophysical finding about those modalities.

14. Scientific boundaries and revised claims

The project can demonstrate properties of explicitly implemented functions: deterministic equality, sensitivity to context, learned task representations, coordinate dependence, temporal recurrence, and coarse output agreement under hidden-state variation. It cannot show whether these programs are conscious, whether two people have the same or different qualia, or which physical process is identical with experience.

The phrase "everything observable agrees" in the iteration-4 conversation was too broad for a report/action test. This app uses "matching selected report and action" and optionally tests a probability tolerance. Other outputs, interventions, or future stimuli can reveal differences. A matched pair on one trial is not a pair equivalent on all possible experiments.

The conversation's statement that physical identity forces a choice between identical experience and a nonphysical difference is a philosophical framing, not an empirical result. The simulator assumes deterministic computational dynamics. It does not test physicalism, dualism, functionalism, illusionism, or any complete theory of consciousness.

The distinction between explaining cognitive functions and explaining felt experience is developed in David Chalmers (1995), Facing Up to the Problem of Consciousness. It motivates this project's boundary between computational outcomes and phenomenal claims; it is not treated as a settled solution. Source: <https://consc.net/papers/facing.html>

Color experience cannot be reduced here to wavelength; the app has no spectra, cone fundamentals, retinal adaptation, calibrated display measurements, or realistic surface rendering. Its single red/green axis is especially limited. The other sensory axes are equally simplified. Statements about real perception require additional models and empirical evidence.

15. Future research, distinct from completed work

The proposed next step is to compare the geometry of complete representational spaces using many stimuli and carefully chosen invariances. Candidate methods include pairwise representational dissimilarity matrices, orthogonal Procrustes baselines, and linear CKA. A richer battery should compare all outputs, confidence, discrimination, generalization, memory, and future-sequence responses.

Use controlled transformations with known ground truth, multiple random seeds, matched training distributions, held-out histories, and uncertainty estimates. Predeclare matching criteria and report nonmatches as well as selected winners. Test how conclusions change when the alignment family changes. Only then consider more realistic sensory encoders or empirical human comparisons with appropriate methodology.

16. Files, maintenance, and practical limits

The app source is `dist/index.html`, `dist/style.css`, `dist/engine.js`, and `dist/app.js`. The engine contains seeded sampling, neural training, transparent features, forward evaluation, and affine alignment. The interface handles experiment state, asynchronous batch search, charts, ablations, temporal updates, and session export/import. The research documents are available at the project root and copied into `dist` for the download links.

The browser performs all work in memory. Training and search yield between small batches so the UI can show progress; the controls are locked while those jobs run. There is no automatic durable session storage, multi-user collaboration, large model training, or hardware stimulus delivery. Session import accepts at most 8 MB and validates core numeric dimensions and categories. A read-only WebMCP comparison tool is registered only where the browser supports `document.modelContext`.

The interface offers visual inspection of eight components rather than pretending a two-dimensional plot preserves the entire state. The history strip shows the last 20 steps, while the exported sequence retains the complete session log. Reset identical preserves the current raw stimulus and selected model type; Clear sequence affects only temporal history. Refresh starts a new default session.

17. Monty integration: a shared sensorimotor protocol

This extension responds to the request: "can we also integrate `tbp.monty` to actually have the touch sensation incorporated, and use the same sensory modality technique for all other stimuli as well, because brain processes any stimuli in the same way".

We operationalize the request as learning from touch-related observations and using a common representation interface across sensory modalities. Incorporating touch data is different from producing tactile feelings. A common computational principle is also different from the claim that all biological sensory systems process input identically. This project tests the former without assuming the latter.

The initial consolidated lab remains available unchanged in its original four panels. A fifth panel, Sensorimotor lab, now runs real `tbp.monty` graph learning on the Mac mini. The original browser-only experiments remain usable offline. This new experiment needs the hosted app and its configured Python runtime; its result is not a browser simulation of Monty.

What is imported, and what is custom

The integration imports Monty's unmodified `State` and `EvidenceGraphLM` classes from the user's `merolaagi/tbp.monty` fork at commit `ed67dcbacdded0418b59ff47667e59159116aa5`. `State` implements feature-at-pose messages. `EvidenceGraphLM` learns object models and accumulates evidence about object and pose hypotheses. If this backend is unavailable, the interface reports a failure; it never substitutes the original toy network and calls it Monty.

The surface environment, prescribed scan path, feature generation, two-agent experiment orchestration, API, and browser interface are new Qualia Lab code. No Habitat or TACTO simulator is used. No physical tactile sensor, skin mechanics, force-feedback device, microphone, camera, taste sensor, or scent sensor is connected. Thus this is a genuine Monty integration with synthetic observations, not hardware integration or a physical haptics system.

Source code: <https://github.com/merolaagi/tbp.monty/tree/ed67dcbacdded0418b59ff47667e59159116aa5>

Monty's cortical messaging protocol is designed to let sensor and learning modules exchange features with locations and orientations. Its architecture motivates the shared interface used here. It does not validate our invented sensory encoders. Reference: <https://docs.thousandbrains.org/docs/cortical-messaging-protocol>

Monty's application criteria emphasize sensorimotor information and warn that it is not simply a drop-in learner for arbitrary static data. Our scan supplies spatial information explicitly. Reference:

<https://docs.thousandbrains.org/docs/application-criteria>

The five adapters

Touch supplies normalized pressure, roughness, and temperature at contact locations. A dimensionless drive parameter affects pressure. The geometric path lies on an analytical sphere, ellipsoid, or radially rippled surface; sensor orientation is supplied from the analytical geometry. This is not a validated contact-force or thermoreceptor model.

Vision supplies synthetic luminance, redness, and contrast. Sound supplies normalized frequency, amplitude, and spectral brightness, with drive affecting amplitude. Taste supplies sweetness, sourness, and bitterness. Smell supplies floral, woody, and intensity features. Each front end has a different explicit transfer function while sending the same structural message type to the same learning algorithm.

All feature values are dimensionless and clipped to [0,1]. There are no hertz, pascals, degrees Celsius, chemical concentrations, or spectral measurements behind these numbers. Sound, taste, and smell reuse virtual spatial poses from the test environment. Whether a realistic sensor for those modalities can provide appropriate spatial information is a research question; the adapter does not solve localization.

Selecting a modality runs separate A/B learning modules for that modality. This version does not combine all five simultaneously, implement native lateral voting between them, or establish cross-modal recognition transfer. It supplies a common implementation path so those future experiments can be designed explicitly.

Learning and test protocol

Agent A learns all three named objects from 36 noiseless contacts per object, using baseline drive 0.5. Agent B either receives identical training, receives the same locations with each feature increased by 0.22 and clipped, or receives only the first half of the training surface. Object labels supervise graph names during training. The inference target is unlabeled; the true test object is used only to generate the test stream and is not given to the matcher as its object label.

The user selects 12, 24, or 36 test contacts, a modality, object, B history, drive, feature noise, seed, and optional intervention. The test scan has an angular offset of 0.037 radians from the training scan, so it is not a literal replay of those positions. This is still a small, highly structured synthetic recognition task, not a benchmark of real-world generalization.

Both agents receive independent copies of the same test observation. The module sees the actual State location, orientation, on-object marker, and selected features. The same EvidenceGraphLM settings are used across the modalities: maximum spatial match distance 0.018 m, per-feature tolerance 0.22, feature weight 1, a 0.3 m graph extent, 50 voxels per dimension, and at most 128 nodes per graph. A single zero-angle initial rotation prior is supplied. Arbitrary-orientation recognition is not evaluated by this protocol.

The scan is prescribed by Qualia Lab. Although location changes provide sensorimotor information, Monty does not choose the movement in this version; no Monty motor policy or goal-state generator is active. Calling this fully autonomous active touch would overstate the integration.

Results and interpretation

The display exposes the most-likely object, candidate object set, number of pose/location hypotheses, learned node count, and maximum evidence per named object after every contact. Evidence is unnormalized and may be negative. The chart uses solid lines for A and dashed lines for B. These are neither class probabilities nor measures of consciousness.

Evidence RMS difference compares A and B across the same three named object scores. It is not the original lab's eight-dimensional hidden-state RMS, and there is no assertion that the two measures share a meaningful scale. Matching most-likely objects does not imply matching full evidence, pose hypotheses, or future behavior.

Export full experiment saves the protocol, pinned source revision, graph-node locations, every test location and orientation, sensory features, hypothesis counts, candidate sets, evidence traces, and boundary statements. The browser's replay slider inspects those returned values; it does not regenerate the experiment while scrubbing.

Use sensor sample in studio maps selected feature values into the old 16-channel toy experiment. For example, touch roughness and temperature become texture and temperature inputs. This is an explicit convenience bridge. It does not transform Monty's graph state into the original network's hidden units, and pressure is not represented as a new original-network channel.

Interventions and controls

The identical-history control should produce exactly equal evidence traces. Biased-feature training tests whether changing the learned sensory associations alters evidence under the same test stream. Partial exploration tests the effect of incomplete graph coverage. These are within-model comparisons.

Zero sensory features removes the three non-morphological feature values at test time while keeping the supplied locations and frames. Freeze reported location gives every test observation the same zero location while keeping the varying frames and features. It tests the contribution of translational information, not the complete removal of pose or embodiment. Neither intervention retrains the object graphs, and both can be outside the training distribution.

Noise is independent zero-mean Gaussian perturbation of the three feature channels before clipping. It is not pose noise or a biological noise model. The seed determines these draws. Contact/acoustic drive affects only the explicitly defined touch and sound features; changing it for vision, taste, or smell has no effect by design.

Runtime and reproducibility

The existing local Monty Python environment is reused without modifying its installed packages. Missing pure-Python dependencies are installed in a project-local overlay. The tested core runtime uses Python 3.8, NumPy 1.23.5, and Torch 1.11.0; it differs from the fork's full recommended dependency versions. Successful integration checks apply to this graph-learning path only, not the optional simulators or every upstream feature.

The public API accepts only the bounded experiment parameters. It runs one experiment process at a time, limits requests to 4 KB and responses to 2 MB, and stops a run after 90 seconds. The UI can receive a busy response when another visitor is running an experiment. Each request trains fresh graphs; there is no upload endpoint or durable server-side experiment memory. The original offline app and session export still operate independently.

For future iterations, preserve the same GitHub repository and domain. Computational tests run before publication; the real Monty integration suite runs on a configured Mac. Backend code changes require restarting the Qualia Lab server in addition to switching the static release. The setup and test commands are documented in `backend/README.md` and `DEPLOYMENT.md`.

Next experiments

A useful next step is actual calibrated tactile data with measured sensor pose, or a separately validated contact simulator. Add a real sensor adapter only after specifying units, calibration, sampling times, missing-data behavior, and pose estimation. For multisensory studies, define which quantities are genuinely shared across modalities and which are not, then test transfer and voting on held-out objects and trajectories.

The neuroscience motivation remains a hypothesis about reusable cortical computation. It is not evidence that touch, vision, sound, taste, and smell are identical, nor that any trained model has subjective sensations.

Appendix A. Recovered conversation transcript

This appendix preserves all returned message text in chronological order. Speech disfluencies, very short acknowledgments, and unfinished responses are retained. Generated-artifact placeholders are marked unavailable because the files were not returned. The scientific qualifications above apply to the early statements below; the transcript is an archival record, not an endorsement of every claim.

Turn 1

User

Okay, uh you and I can talk about uh Consciousness, are you ready

Assistant

Absolutely. Consciousness is a big topic. Where do you wanna start?

Turn 2

User

Okay, so uh You and my, you imagine a red apple and uh red uh ball So what is the difference between the redness of red apple and redness of a red ball

Assistant

Hmm, interesting. So, the red of an apple often feels like it has depth. There's that waxy skin with subtle variations, maybe tiny speckles. The red of a ball, say a rubber ball, tends to be more uniform, kind of flat, even synthetic. So, even if they share a similar wavelength of red light, our experience of them can be very different. Do you wanna go towards how that ties into subjective experience? Like qualia?

Turn 3

User

That's what I was saying, why do you feel the

Assistant

Mm.

Turn 4

User

Redness should be same because your brain is perceiving the same light as that

Assistant

That makes sense on paper, but our experience of color isn't just wavelength, it's context, surface texture, lighting, associations. So, that redness is being filtered through all that.

Turn 5

User

So, ah Then uh, with a two people So the redness the redness experienced by you and redness experienced by me is same or different? Different right

Assistant

That's a classic. We can't directly compare what my red feels like versus yours. We assume it's similar because we agree on labels and behavior, but the actual private feel, that subjective bit, we can't swap that. That's part of what makes consciousness so slippery.

Turn 6

User

Okay let's try to tackle this one so Whoa. What do you think comprises the the subjective this qualia, what what What is it actually

Assistant

One way to look at this is to distinguish between the physical story and the felt story. Physically, it's light hitting your retina, neurons firing, information processing. But the felt side, the what it's like part, that's the part we don't yet know how to pin down. Whether it's identical to some physical process, or emerges from complex information integration, or needs some new kind of explanation, is still an open question. That's basically the hard problem of consciousness. And with the redness comparison, we just don't have a way to line up two subjective feels side by side in a lab and check for sameness. The privacy is built in. Maybe a useful next step is asking whether we need to posit something extra at all, or if subjective feel is just what certain kinds of processing feel like from the inside. Does that framing click for you, or does it still leave something

Turn 7

User

No, okay

Assistant

Hmm.

Turn 8

User

Okay okay so let me let me let me create a hypothetical situation Let's try to update inside a subjective feeling by creating two identical objects identical being. Uh Identity- Identical in all all aspects, so The same red

Assistant

Okay, let me check that. Imagine you build two beings, A and B, identical down to the last neuron and state, seeing the same red stimulus. Now the question is, does perfect physical identity guarantee identical qualia? If you say yes, you're treating experience as fully fixed by the physical; if you say no, then you're implying there's some difference with no physical footprint. Your hypothetical forces that fork. Either identical structure entails identical experience, or you accept some non-physical difference. That's why this is such a tough problem. There isn't an experiment that cleanly answers it yet. Does that line up with where you were trying to go, or were you aiming at a different angle?

Turn 9

User

Okay. Why don't- Okay, why don't we, for example, create a little uh- little uh- uh little app where we have two uh Two uh, two uh uh subject and we'll try to objectify the redness uh

Assistant

Hmm.

Turn 10

User

. Give them give them the same stimuli and then see uh uh- What uh- because it has to be the same thing. And at one point let's give them, let's randomize their uh subjective experience. I don't know if the if the randomization comes into play or something that the subject the objective feeling suddenly becomes a subjective experience

Assistant

Hmm. I think your app idea, used carefully, can help clarify what can and can't be shown. Picture two simulated subjects, A and B. You feed them the same input, say, 650 nanometer light. Inside each subject, there's some internal state we call Q, their red experience state. Externally, you only ever observe reports or behavior. You could then ask if we secretly remap B's Q, swapping internals while keeping all outputs identical, would anyone ever know? That's basically an inverted spectrum test. Now, if you add your randomization idea, varying Q while measuring everything reportable, then you can ask: At what point does a change in Q show up in function like memory, comparison, or report? One ironic upshot: If nothing changes anywhere in the system when you tweak Q, then you're just labeling something as subjective without operational grip; but if something does change, you've made it measurable in principle. So the app might not solve qualia, but it could be a neat thought-experiment simulator to test where subjective claims become objective. Does that get at your angle?

Turn 11

User

Okay, yeah, that is actually what I want to build and also the second thing is like uh At one point, say for example

Assistant

Huh.

Turn 12

User

the the the the the sensory, the other sensory modalities might come into picture how it feels, what was the context when the red apple was seen or red ball is seen You know, uh, what was the environment like? How much shadow uh that particular red ball had or some other thing had. Mix all those experience or test to to that particular stimulus. and uh Maybe the angle of the light. Everything, everything possible. Like you know may- and then uh, we we kind of like, you know, test each one uh in two separate context, add same experience, subtract one experience, play with all those, play mix them up and then see what actually changes uh the behavior. Do you know what I'm trying to say

Assistant

Um.

Assistant

Yes, I think I get the shape of it. What you're sketching is not well, just a red test anymore. It's a parameter sweep over experience itself. Start with identical agents and define an experience vector that includes color plus context, shadow, texture, smell, prior memory, emotional tag, et cetera. Then systematically vary one dimension

Turn 13

User

Based on our conversation so far let's try to build a small app

Assistant

Yes. I built a first small working prototype based directly on our consciousness discussion:

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It gives us **Subject A and Subject B** and lets us manipulate things like redness, saturation, illumination, shadow, context, prior memory, emotional association, and influence from other senses.

There are also three experiments I specifically wanted in this first version: make A and B identical, randomize B's context while preserving the stimulus, and **invert B's private color-label mapping**-our version of the inverted-spectrum thought experiment.

Importantly, the app distinguishes **observable behavioral reports** from qualia itself. It doesn't pretend that a numerical score is consciousness. That's actually the scientific problem we're trying to investigate.

The next version could become much more interesting: instead of me arbitrarily defining how these variables combine, we can create two little artificial agents with **sensory state -> memory -> internal representation -> report/behavior**, and systematically add/subtract experiences one at a time. That gets much closer to the experiment you were describing.

Turn 14

User

Yes let's do it

Assistant

Done - I built **Version 2** as the two-agent experiment we were describing.

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This version models:

Stimulus -> sensory processing -> context -> memory -> internal representation -> report/behavior

Start with **Reset: identical agents**. Then change only Agent B. You can subtract memory, context, other senses, or emotion; randomize B's experiential history; or invert B's internal color mapping.

The particularly interesting experiment is **Invert B internal mapping**. We can now investigate whether two agents could produce similar observable behavior while having systematically different internal representations.

For Version 3, I'd like to push your idea further: **don't hand-code what "redness" means at all**. Give each agent a small learnable network, expose them to different histories, and see whether an internal representation of red *emerges*. Then we can inspect the internal geometry and ask whether A's "red" and B's "red" are actually equivalent representations. That would turn the philosophical thought experiment into a small computational experiment.

Turn 15

User

Sure and also with each iteration, can we have the idea in readme.md or pdf format

Assistant

Absolutely. From now on, each iteration will include the **working app + updated README.md + PDF research note**, so we preserve not only the code but the evolution of our thinking.

For the current iteration:

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I also documented an important principle for the project: **we should not simply label an internal variable "qualia."** We'll call it an *internal representation* and investigate when it diverges from observable behavior.

For **Iteration 3**, we'll move away from our hand-written formula for redness. The two agents will **learn their own internal representation from experience**. Then we can give them identical versus different experiential histories and examine whether their representations of the *same red* converge or diverge.

That's where this starts becoming a genuine computational experiment rather than only a visualization.

Turn 16

User

Ok let's do the third iteration

Assistant

Iteration 3 is ready. This is a meaningful jump from the previous version.

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What's new

We no longer explicitly tell the agents what "redness" is. Each agent now develops an **8-dimensional latent representation through learning**.

You can run three particularly useful experiments:

1. **Identical histories** - A and B experience the same training data.
2. **Different histories** - the same color categories, but B develops them under different context, memory, and multisensory associations.
3. **Inverted labels** - B learns a different language mapping for red/green.

Then give both agents **exactly the same red stimulus** and compare their learned internal representations.

I also added an **alignment experiment**. This matters because two brains-or two neural networks-could represent the same information in different internal coordinate systems. Neuron/unit #3 in A doesn't necessarily have to correspond to unit #3 in B.

So our question is becoming more precise:

> If A and B both say "red," what properties of their internal representations actually have to be the same?

And an even more interesting target for Iteration 4 is now apparent: have the program automatically search thousands of conditions for cases where **A and B behave identically while their internal representations are maximally different**.

That would directly attack the original question that started our conversation.

Turn 17

User

Ok let's do iteration 4

Assistant

Iteration 4 is ready.

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The big experiment in Iteration 4

The app can now generate **thousands of experiences automatically** and specifically search for this condition:

> Agent A behavior = Agent B behavior, but Agent A internal representation != Agent B internal representation.

So instead of manually finding interesting examples, click "**Search same-behavior / different-state cases.**" It searches 1,000-15,000 cases and finds the one with the largest internal divergence despite matching report and action.

We also added **temporal memory**. What an agent experienced immediately before seeing the red object can now influence its current representation.

And once we discover a strong case, we can perform **ablations**: remove memory, context, other sensory information, or object shape and ask which removal makes the two internal states converge.

This gives us a more precise version of our original philosophical problem:

If everything observable about two subjects agrees, how much can their hidden internal representations still differ?

Iteration 5 can take the next important step: instead of looking at individual cases, we'll measure the **geometry of the entire representational space** of A versus B and test whether their apparently different internal worlds are genuinely structurally different-or merely the same representation expressed in different internal coordinates.

Turn 18

User

give me everything in one app and all the conversation we had in single pdf or readme

Assistant

Yes. I've moved this into **Work mode** so it can build the consolidated project properly.

The deliverable will contain:

- **One unified Qualia Lab app** combining Iterations 1-4.
- **One master README.md** capturing the complete evolution of the idea.
- **One master PDF** preserving our discussion from the original *red apple vs. red ball* question through identical observers, qualia, context, memory, multisensory experience, learned representations, inverted mappings, behavioral twins, ablations, and representational divergence.
- Clear separation between **our hypotheses**, **what the simulation demonstrates**, and **what we cannot yet claim about consciousness**.

- The full iteration history and suggested next research direction.

I also asked it to preserve the reasoning behind the project, rather than merely merging the four apps.

Appendix B. Consolidation request and generalization

The current request asked for one consolidated Qualia Lab app containing iterations 1-4, one comprehensive README, and one PDF preserving the evolution from the initial red apple versus red ball discussion. The subsequent steering asked to continue making a high-end, general version rather than limiting the intent to describing redness qualia. The generalized edition implements that direction through five abstract sensory domains while retaining the original redness case study.

This appendix summarizes the current instructions; Appendix A is the recovered source conversation.